Sanjeev Kumar Singh

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Summary

Full stack engineer with 2+ years shipping production systems end to end — Python/FastAPI services, AWS serverless pipelines, cross-platform Flutter apps on both app stores, and applied ML from dataset pipeline to trained model. Comfortable owning a feature across data, business, and API layers.

Experience

AlignTogether Solutions Pvt. Ltd. October 2024 – Present
Software Development Engineer - II *Bhopal, M.P*

Hexalog — OMS & Reporting Layer | *Python, FastAPI, PostgreSQL, SQLAlchemy, AWS S3*

- Owned a document-verification feature end to end across the data, business, and API layers of a multi-tenant FastAPI platform, with a state machine guarding removal, approval, and entity-type transitions.
- Built two-way sync with the ShipThis freight ERP using version watermarking to avoid re-sending documents, plus inbound webhooks for verification verdicts and ops-uploaded files.
- Built email subscription infrastructure enforcing preferences at queue time, with RFC 8058 one-click unsubscribe headers, audit logging, and Alembic migrations across 3 environments.

Edept — Odoo & Automations | *Python, FastAPI, Odoo 19, PostgreSQL, WebSockets*

- Built a FastAPI service bridging live Plivo telephony audio with ElevenLabs conversational AI over WebSockets, with a concurrency-capped dispatch queue, dedupe, and working-hours-aware retry scheduling.
- Replaced ORM-based lead import with direct transactional SQL, cutting multi-hour imports of 50K+ rows to seconds, with Postgres-native dedupe and a SAVEPOINT per row so one bad row cannot fail the batch.
- Built durable-queue services syncing Ozonetel CDRs idempotently by UID and delivering CRM events to the Meta Conversions API in batches of 1000, with SHA-256 hashed PII, backoff retry, and dead-lettering.
- Shipped 10+ Odoo modules covering lead routing, deduplication, click-to-call CTI, and SQL-view-backed admissions dashboards.

Kubocare — Radar Pose Estimation | *Python, TensorFlow, Keras, MQTT, ExPose*

- Researched and implemented mmWave radar point cloud to human skeleton prediction for posture recognition.
- Built a pipeline generating a ˜150K-sample dataset from raw radar recordings, with 3D mesh and skeleton labels from ExPose for supervised training.
- Trained a TensorFlow/Keras model predicting skeletal joints, and streamed live radar over MQTT to visualise real-time skeletal motion.

ShePower — Social Platform App | *Flutter, Dart, Node.js, Express, MongoDB, Azure*

- Built and published a Flutter social application to both the Play Store and App Store, owning the Android and iOS release process across 16 versioned releases.
- Implemented authentication with OTP verification and role-based registration flows.
- Migrated backend storage from AWS S3 to Azure Blob with a compression pipeline; added WebSocket events, notification polling, and Razorpay payments.

Value First — Surbo Chat | *Flutter, Dart, BLoC, Socket.IO, Twilio SDK, Firebase*

- Built and shipped a cross-platform Flutter chat application to both the Play Store and App Store, as primary contributor on the codebase.
- Implemented real-time messaging over Socket.IO with media, delivery receipts, and role handling, and integrated native Twilio audio and video calling for Android and iOS.
- Integrated 100+ backend APIs covering chat, user management, and application features.

Lakshhr — Serverless Automation | *Node.js, Python, AWS Lambda, SQS, SES, PostgreSQL*

- Migrated 30+ business workflows from Airtable to PostgreSQL with zero data loss.
- Rebuilt them on an event-driven AWS serverless architecture using Lambda, EventBridge, SQS, and SES, with queue-based messaging and automated transactional email.
- Developed the Node.js backend managing APIs, data pipelines, and automation logic.

Fittr — Fittr Hart | *Flutter, Dart, Matrix SDK, BLoC, Realm, Firebase*

- Developed modern, responsive mobile UI in Flutter with a clean and maintainable architecture.
- Integrated Matrix SDK to enable secure real-time messaging, building a full-featured chat module.
- Integrated Realm database for efficient local storage and offline data synchronisation.

Dalmia — CSR | *Flutter, Dart, REST APIs, MySQL*

- Built an offline-capable Flutter Android application for field household surveys and beneficiary enrolment.
- Implemented draft saving, validation, and conditional multi-step form flows for low-connectivity field use.
- Shipped through a DEV/QA/production release flow across 60+ tagged versions.

Affirmity Corp December 2023 – September 2024
Software Development Engineer Intern *Bhopal, M.P*

Perimeter Security System | *Flutter, Python, Flask, OpenCV, YOLOv8, FaceNet*

- Built a multi-window, multi-tab Flutter desktop application with a Flask backend for real-time surveillance.
- Used custom-trained YOLOv8 for object detection and FaceNet for high-accuracy face recognition, extended to drone video analysis with OpenCV and MTCNN.
- Enabled fast event filtering and querying over detection data using Elasticsearch and MySQL.

Projects

PlanMyTrip | *Next.js, FastAPI, PostgreSQL, AWS S3, Razorpay* *Link*

- Full-stack group travel platform: itineraries, expense splitting, shared albums, and payment collection.
- Razorpay payments with signature-verified idempotent webhooks, and direct-to-S3 uploads with background thumbnailing.
- Live activity feed over WebSockets and scheduled email reminders through SES.

AuthForge | *FastAPI, React, Flutter, PostgreSQL, Redis, OAuth2* *Link*

- Self-hosted identity platform with TOTP multi-factor auth, session and trusted-device management.
- Full OAuth2 + PKCE identity provider, so other applications can "Sign in with AuthForge".
- Includes a Flutter authenticator app generating RFC 6238-compliant offline TOTP codes.

MyPassword | *Next.js, TypeScript, Firebase, AES-256-GCM, Argon2id* *Link*

- Zero-knowledge password vault where all cryptography runs in the browser — keys never reach the server.
- Envelope encryption with AES-256-GCM data keys and Argon2id key derivation running in WebAssembly.
- Auto-locking after inactivity and full re-encryption on master password rotation.

Win Rewrite | *AutoHotkey v2, Lua, Bash, Google Gemini API* *Link*

- Cross-platform desktop utility rewriting selected text in place via the Gemini API with a single hotkey.
- Works in any application while preserving the original clipboard contents.
- Runs in the system tray with a configurable shortcut and an automated Windows installer.

BlogGen | *Next.js, TypeScript, Firebase, Firestore, Tailwind CSS* *Link*

- Full-stack blogging platform for writing, managing, and publishing posts.
- Content stored in Firestore with security-rule-enforced access and Google authentication for authors.
- Responsive UI built with Tailwind CSS and shadcn/ui, deployed on Vercel.

Trimly | *Next.js, MongoDB, Server Actions, App Router* *Link*

- Secure and scalable URL shortener with an interactive, responsive UI.
- Built on the Next.js App Router with Server Actions for optimised server-side execution.
- MongoDB primary store for reliable link storage and high-speed lookups.

Technical Skills

Languages: Python, Dart, TypeScript, JavaScript, SQL
Backend: FastAPI, Flask, Node.js, Express.js, Odoo, SQLAlchemy, REST APIs, WebSockets
Frontend & Mobile: Flutter, BLoC, Next.js, React, Tailwind CSS, HTML/CSS
AI/ML: PyTorch, TensorFlow, Keras, OpenCV, YOLOv8, Scikit-learn, Pandas, NumPy, FFmpeg
Cloud & DevOps: AWS (Lambda, EventBridge, SQS, SES, S3, EC2, SageMaker), Docker, Kafka, MQTT, CI/CD
Databases: PostgreSQL, MySQL, MongoDB, SQLite, Redis, Realm, Hive

Education

Sagar Institute of Science and Technology Bhopal, M.P
B.Tech in Computer Science and Engineering (CGPA: 8.5) *Oct 2020 – May 2024*